

# Exercise 2:

Code:

Document.java:

```
1  public interface Document {  
2      void open();  
3      void save();  
4  }
```

WordDocument.java:

```
1  public class WordDocument implements Document {  
2      @Override  
3      public void open() {  
4          System.out.println(x: "Opening Word document...");  
5      }  
6  
7      @Override  
8      public void save() {  
9          System.out.println(x: "Saving Word document...");  
10     }  
11 }
```

PdfDocument.java:

```
1  public class PdfDocument implements Document {  
2      @Override  
3      public void open() {  
4          System.out.println(x: "Opening PDF document...");  
5      }  
6  
7      @Override  
8      public void save() {  
9          System.out.println(x: "Saving PDF document...");  
10     }  
11 }
```

ExcelDocument.java:

```
1  public class ExcelDocument implements Document {
2      @Override
3      public void open() {
4          System.out.println(x: "Opening Excel document...");
5      }
6
7      @Override
8      public void save() {
9          System.out.println(x: "Saving Excel document...");
10     }
11 }
```

DocumentFactory.java:

```
1  public abstract class DocumentFactory {
2      public abstract Document createDocument();
3  }
4  
```

WordDocumentFactory.java:

```
1  public class WordDocumentFactory extends DocumentFactory {
2      @Override
3      public Document createDocument() {
4          return new WordDocument();
5      }
6  }
```

PdfDocumentFactory.java:

```
1  public class PdfDocumentFactory extends DocumentFactory {
2      @Override
3      public Document createDocument() {
4          return new PdfDocument();
5      }
6  }
```

ExcelDocumentFactory.java:

```
1 public class ExcelDocumentFactory extends DocumentFactory {
2     @Override
3     public Document createDocument() {
4         return new ExcelDocument();
5     }
6 }
```

FactoryMethodTest.java:

```
1 public class FactoryMethodTest {
2     public static void main(String[] args) throws Exception {
3         System.out.println("Testing Factory Method Pattern Example:");
4         DocumentFactory wordFactory = new WordDocumentFactory();
5         Document doc1 = wordFactory.createDocument();
6         doc1.open();
7         doc1.save();
8         System.out.println();
9
10        DocumentFactory pdfFactory = new PdfDocumentFactory();
11        Document doc2 = pdfFactory.createDocument();
12        doc2.open();
13        doc2.save();
14        System.out.println();
15
16        DocumentFactory excelFactory = new ExcelDocumentFactory();
17        Document doc3 = excelFactory.createDocument();
18        doc3.open();
19        doc3.save();
20    }
21 }
```

## Output:

```
Testing Factory Method Pattern Example:
```

```
Opening Word document...
```

```
Saving Word document...
```

```
Opening PDF document...
```

```
Saving PDF document...
```

```
Opening Excel document...
```

```
Saving Excel document...
```